Waveshare Wiring - Case Feeder Pro

Waveshare ESP32-S3-Touch-LCD-2.8

Firmware env `lcd_waveshare`, for a V1 board and for a V2 / Rev1.1 board. The cable is the same either way. On the SKR Pico the plug is the **Raspberry Pi UART** connector, the 5-pin JST-XH header printed `5V 5V GND IO1 IO0`. One of the two `5V` pins is used; the other stays empty.

Waveshare's wiki and the schematic linked from that page name the plugs:

- The **UART header** is a **4-pin SH1.0** (1.0 mm) connector: `GND`, `3V3`, `TXD` (GPIO43), `RXD` (GPIO44). It has no 5 V pin.
- The **12-pin header** along the bottom edge is **SH1.0-12P**. `VBus` on that header is the USB 5 V rail.
- The **speaker header** is a **1.25 mm 4-pin** right-angle connector on the amplifier outputs. The wiki supplies an 8 Ω 2 W 2030 speaker for it.

```
SKR Pico, Raspberry Pi UART          Waveshare
5V  ------------------------------>  VBus on the 12-pin header
5V  (the other 5V pin stays empty)
GND ------------------------------>  GND on the UART header
IO1 (Pico RX) ------------------->  TXD  GPIO43
IO0 (Pico TX) ------------------->  RXD  GPIO44
                                     3V3 on the UART header stays open
```

SKR Pico Raspberry Pi UART	Waveshare
`5V`	`VBus` on the 12-pin header
`GND`	`GND` on the 4-pin UART header
`IO0` (Pico TX)	`RXD` (GPIO44)
`IO1` (Pico RX)	`TXD` (GPIO43)
leave open	`3V3` on the UART header

- `3V3` on the UART header is an output of the onboard regulator. Leave it unconnected. Pico `5V` goes to `VBus` only, never to a pin labeled `3V3`.
- Use **one ground**: UART `GND` to Pico `GND`. Do not add a second ground wire to the 12-pin header.
- Do not feed Pico `5V` into `VBus` while a computer USB-C cable is plugged into the Waveshare. `VBus` is that USB 5 V rail. Unplug the Pico `5V` wire before flashing the board over USB.
- If the screen shows **NO LINK**, swap the two data wires.
- Fit a **470 Ω resistor in series with each data wire**, twist each data wire with the ground wire, keep the cable short, and clip a ferrite core on near the display. Same as the ES3C28P cable.
- The separate 4-pin I2C header (also SH1.0) is the IMU and RTC bus. Leave it unconnected.

The speaker plugs into the speaker header. The knob on the board sets the hardware volume; Settings > Volume still scales the tones in software.

The board is 73.06 × 50.54 mm. Its connectors are not in the same places as the ES3C28P, and the printed lid was not drawn for it. Measure the board before committing to that lid.

The label on the back says which touch chip is fitted. V1 is the CST328. V2, or a board silkscreened Rev1.1, is the CST3530. Display, backlight, speaker, and this cable are the same on both. The USB serial line names the chip it found. A CST3530 reset takes about half a second, so the first picture is a little later on a V2 board.

Source: Infringer13/Case-Feeder-Pro-Waveshare docs/BUILD_GUIDE.md @ a697d1e